

Response to the Reviewers

Manuscript: “Trust-Aware Multi-Agent Reinforcement Learning-Based Cooperative Interception Strategy for Maneuvering Swarms with Target Assignment”

We thank the reviewer for the careful reading and the constructive comments. Below, each comment is reproduced verbatim (grey box), followed by our point-by-point response, the exact location and wording of the corresponding revision in the manuscript (blue box, matching the blue highlight in the revised PDF), and the supporting figures prepared for this letter. Equation and section numbers refer to the revised manuscript.

Reviewer 1

Comment 1.1

Reviewer comment 1.1

Provide a concise notation table summarizing key variables and symbols used throughout the paper.

Response. We thank the reviewer for this constructive suggestion. We have added a dedicated notation appendix (**Appendix A**, “Summary of Key Notation”) to the revised manuscript. It consolidates the principal variables and symbols into a single table, organized by the four themes of the paper: the engagement geometry and kinematics, the IDBO-based target assignment, the ART-MAPPO cooperative guidance, and the simulation/evaluation metrics. Symbols that recur across contexts are disambiguated by their subscripts (for example, the heading angle γ versus the reinforcement-learning discount γ_{RL} , and the assignment weights w_1, w_2 versus the reward-component weights $w_1, \dots, w_5$).

Comment 1.2

Reviewer comment 1.2

Include pseudocode or algorithm boxes for IDBO and ART-MAPPO to improve reproducibility.

Response. We thank the reviewer for this helpful suggestion. We have added two formal algorithm boxes to the manuscript: **Algorithm 1** (Distributed Consensus-Based IDBO) at the end of Section III-B, and **Algorithm 2** (ART-MAPPO training under CTDE) at the end of Section IV-D. Because the comment explicitly targets reproducibility, these boxes are added to the *main paper* (highlighted in blue in the revised manuscript). Crucially, *every line of both algorithms is cross-referenced to the equation that defines it*, so each step can be reproduced directly from the corresponding formula. Both boxes use the `algorithm/algorithmic` environments already provided by the IEEEtran template; no equations were modified, the manuscript still compiles cleanly, and all equation cross-references inside the boxes resolve correctly.

Location of the revision. Section III-B, immediately after Eq. (30) (Algorithm 1); Section IV-D, immediately after the ART-MAPPO workflow figure (Algorithm 2). For the reviewer’s convenience the two boxes as they now appear in the manuscript are reproduced below.

Algorithm 1 Distributed Consensus-Based IDBO for Cooperative Target Assignment (per interceptor u_i)

Require: Interceptor set $\mathcal{U}$, target set $\mathcal{T}$, probability matrix $\mathbf{P}$ (12), neighbors $\mathcal{N}_i$, capacity $L_{\max}$, IDBO iterations T , population size N_{pop} , threshold ϵ

Ensure: Converged assignment $\mathbf{X}^*$

- 1: Initialize preference population $\{\mathbf{x}_i\}$, winner lists $\mathcal{Z}_j \leftarrow \emptyset$, timestamps $\mathbf{y}_i$, iteration $k \leftarrow 0$
 - 2: **repeat**
 - 3: *// Phase 1: local metaheuristic optimization*
 - 4: **for** $t = 1$ to T **do**
 - 5: Set decaying coefficients $c_t = c_0(1 - t/T)$ for $\{\alpha, \beta, \gamma, \delta, \delta', \eta, \sigma_d\}$
 - 6: Evaluate fitness $\mathcal{F}_i$ (17) and advantage $\mathbf{A}_i$ (19)
 - 7: Update $\mathbf{x}_i$ by Rolling/Dancing/Breeding/Stealing (20)–(23)
 - 8: **end for**
 - 9: Binarize preferences by advantage-adaptive thresholding (24)
 - 10: *// Phase 2: auction-based distributed consensus*
 - 11: Compute bids c_{ij} (25)
 - 12: Exchange winner lists with $\mathcal{N}_i$; update top- $L_{\max}$ set $\mathcal{Z}_j$ (26)
 - 13: *// Phase 3: adversarial-information update*
 - 14: Refresh advantages $\mathbf{A}_i^{(k+1)}$ from the consensus assignment
 - 15: Compute swarm disagreement $\Gamma^{(k)}$ (30); $k \leftarrow k + 1$
 - 16: **until** $\Gamma^{(k)} < \epsilon$
 - 17: **return** consensus assignment $\mathbf{X}^*$
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Algorithm 2 ART-MAPPO Training for Cooperative Swarm Interception (CTDE)

Require: Actor/critic params θ, ϕ ; clip ϵ , dual-clip floor c ; target KL δ_{targ} ; rates α_T, ρ, τ_T ; epochs K

Ensure: Trained decentralized policy π_θ

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1: Initialize  $\theta, \phi$ , trust  $\mathcal{T}_i \leftarrow 0.5$ , KL weight  $\beta_{\text{KL}}$ , running stats  $\mu_R, \sigma_R^2$ 
2: for episode  $k = 1$  to  $K$  do
3:   // Decentralized rollout collection
4:   for each step  $t$  and interceptor  $i$  do
5:     Encode  $\mathbf{o}_i^{(t)}$ ; attention-residual features (32)–(35), temporal state  $\mathbf{c}_t$  (40)
6:     Form policy  $\pi_{\theta_i}$  (41); sample action via trust-blended policy (47), (48) with  $\beta(\mathcal{T}_i) = 1 - \mathcal{T}_i$ 

7:     Clip  $a_i$  to envelope  $\mathcal{A}$ ; step environment; store transition
8:   end for
9:   // Trust update from episode returns
10:  Update  $\mu_R, \sigma_R^2$  (43), (44); normalize  $\tilde{R}_i^{(k)}$  (45)
11:  Smooth trust  $\mathcal{T}_i^{(k+1)}$  (46)
12:  // Centralized optimization (shared critic on global state (42))
13:  Compute advantages  $\hat{A}_t$  and value targets via GAE (50), (51)
14:  for each mini-batch epoch do
15:    Evaluate dual-clip surrogate  $J^{\text{DC}}(\theta)$  (52), (53)
16:    Estimate  $\hat{D}_{\text{KL}}$  (55); value loss  $L_V(\phi)$  (57)
17:    Minimize total loss  $\mathcal{L}(\theta, \phi)$  (58) with AdamW (cosine LR)
18:  end for
19:  Adapt KL weight  $\beta_{\text{KL}}^{(k+1)}$  (56)
20: end for
21: return decentralized policy  $\pi_\theta$ 
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Comment 1.3

Reviewer comment 1.3

Add intuitive explanations before or alongside major equations, especially in Sections III and IV

Response. We appreciate the reviewers’ suggestions. We have supplemented the discussions on convergence statements and equilibrium properties from two aspects: target assignment algorithm and reinforcement learning training. First, a new subsection “Discussion on Convergence and Equilibrium Properties” has been added in Chapter 3, which provides a more rigorous explanation of the convergence conditions and equilibrium properties of the IDBO distributed target assignment process. We clearly state that this convergence discussion addresses the distributed assignment procedure under fixed engagement snapshots, rather than an unconditional proof of global optimality for the nonconvex mixed-integer optimization problem. Under the conditions of a connected communication graph, bounded communication delays, a finite feasible assignment space, and local elite-preserving updates, the local accepted fitness sequence is non-decreasing and eventually reaches a fixed point or a finite plateau. Meanwhile, the top- L_{max} bidding consensus mechanism ensures that each interceptor obtains a consistent winner list. Based on this, we further clarify that the final assignment can be interpreted as a locally meaningful ϵ -approximate Nash equilibrium,

meaning that no interceptor can improve its payoff by more than ϵ through unilaterally changing its target assignment. In addition, we have supplemented the interpretation of convergence for ART-MAPPO in the subsection “RL Model Training and Convergence Analysis” of the simulation chapter. Since neural network policy optimization in multi-agent deep reinforcement learning is a nonconvex problem, the convergence of ART-MAPPO should not be understood as a theoretical guarantee of global optimality, but rather as empirical training convergence. We further explain that the stable training process is supported by the average reward curves across five random seeds, small inter-seed variance, non-divergent critic loss, and gradually decreasing policy entropy. The dual-clip PPO and adaptive KL mechanisms are mainly used to constrain the magnitude of policy updates, reduce policy oscillations, and thus improve training stability. With these revisions, the equilibrium properties of IDBO and the empirical convergence of ART-MAPPO are both defined more clearly and rigorously in the paper.

In the same spirit of aligning each equation with the now three-dimensional engagement, we also updated the angular-advantage term of the target-assignment probability so that its intuitive meaning matches the 3D geometry: the aspect error $S_\sigma(i, j)$ was generalized from a planar heading difference to the Euclidean norm of the azimuth and pitch-plane angular errors, using the same (γ, θ, q) notation introduced in the Section II 3D kinematics. The modified text now reads:

$$S_\sigma(i, j) = \sqrt{(\gamma_{ij} - q_{ij})^2 + (\theta_{ij} - q_{ij}^\theta)^2} \text{ denoting the three-dimensional angular error.}$$

Location of the revision. Section III-A (*Problem Formulation*), in the definition of $S_\sigma(i, j)$ immediately below the angular-advantage equation; the added pitch-plane term is highlighted in blue.

Comment 1.4

Reviewer comment 1.4

Expand discussion on real-world applicability, including communication delays, sensing uncertainty, and 3D dynamics.

Response. We thank the reviewer for this suggestion, which we have addressed structurally rather than as an added paragraph. The revised evaluation is now conducted in a full *three-dimensional* engagement, and the three factors the reviewer lists are built into the closed loop rather than discussed in the abstract:

- **3D dynamics.** The kinematic and line-of-sight models in Section II are extended from the planar formulation to three dimensions. A pitch (flight-path) channel and its associated normal load n_z are added, the velocity is resolved into $(\dot{x}, \dot{y}, \dot{z})$ through the pitch and heading angles, and the zero-effort-miss and relative-geometry relations are rewritten in vector form. The original planar definitions are preserved as the horizontal projection, so the extension supplements rather than replaces the established notation.
- **Communication delay.** A sensing delay of 100 ms and a command-channel delay of 50 ms are injected as explicit delay lines in the evaluation (new Communication-Delay and Observation-Uncertainty table in Section V); its effect on the distributed IDBO consensus is quantified directly below and analyzed in full in Comment 3.2.

- **Sensing uncertainty.** Zero-mean Gaussian position ($\sigma_p = 3$ m) and velocity ($\sigma_v = 0.3$ m/s) errors are added to every observation; a dedicated robustness study (Comment 3.7) quantifies the guidance loop’s tolerance.

To make the communication-delay point concrete, we sweep the per-link delay τ on the paper’s engagement (40 defenders, 12 targets) and measure the time for the distributed consensus to reach and hold full agreement ($\Gamma \rightarrow 0$):

Distributed-consensus time vs. communication delay (40 vs. 12 engagement; nominal 20 Hz link).

Communication delay τ (ms)	0	50	100	200	400	800
Time to consensus (s)	0.57	0.90	1.23	1.88	3.08	5.54

Every setting still reaches full consensus; the delay changes neither the fixed point nor the correctness, only the speed, with the consensus time growing linearly in the delay. A representative 20 Hz link ($\tau = 50$ ms) settles the assignment in under 1 s, so real-world communication delay is tolerated rather than merely discussed. The full complexity, scalability, and diameter analysis is given in Comment 3.2.

These are demonstrated on a hardware-in-the-loop, semi-physical platform (Comment 3.10), so “real-world applicability” is shown empirically.

Location of the revision. Section II (3D kinematics, LOS geometry, loads, and ZEM equations, all highlighted in blue); Section V-A (*Experimental Setup and Implementation Details*): the communication-delay and observation-uncertainty model , the HIL platform description.

Comment 1.5

Reviewer comment 1.5

Discuss the sensitivity of the reward function with respect to weights and hyperparameters.

Response. We thank the reviewer and address the two axes of this sensitivity separately: the *reward weights* of Eq. (66), which shape the learning objective, and the *training hyperparameters* of Table I, which govern the optimizer. Both are now treated in the new sensitivity subsection, Section V-G (*Parameter Sensitivity Analysis*).

1) Reward weights (objective shaping). As detailed in our reply to Comment 3.5 and in Section V-G-2) / Table VII, a one-at-a-time perturbation of the five reward weights w_1 – w_5 —each scaled by $s \in \{0.50, 0.75, 1.00, 1.25, 1.50\}$ with the policy retrained and re-evaluated at every setting—shows that each weight primarily shapes its intended tactical dimension: w_4 governs temporal coordination ($E_{co-time}$ swings $\approx 55\%$), w_5 governs terminal overload E_n , w_1 governs terminal precision E_{miss} , and w_3 governs reliability (ISR). Across the entire sweep the interception success rate stays high ($\geq 96.2\%$ except for the strongly under-weighted terminal-hit case), so the shaping weights are tactical dials rather than knife-edges on effectiveness. We do not restate the full metric table here to avoid duplicating Comment 3.5.

2) Training hyperparameters (optimizer). We conducted a one-at-a-time (OAT) sensitivity study over the three principal optimization hyperparameters—the initial learning rate η_0 , the PPO clipping factor ϵ , and the adaptive-KL target δ_{targ} . Each parameter is swept over five values while the others are held at their nominal settings (listed with their optimization roles in Table I), and the policy is retrained under the same training budget for every configuration. We report the training AUC (area under the normalized-reward curve, in units of 10^4) and the final normalized reward (averaged over the last 500 episodes) in the new Table VIII (manuscript Section V-G-3)). The learning rate has the strongest influence: moving η_0 from the nominal 3×10^{-4} to 1×10^{-3} lowers the training AUC and final reward by 13.9% and 5.8%, whereas the neighboring value 2×10^{-4} changes them by only 3.3% and 0.3%. The clipping factor is essentially flat over $\epsilon \in [0.15, 0.25]$ (variations $\leq 0.60\%$ in AUC and $\leq 0.31\%$ in final reward relative to nominal), and $\delta_{\text{targ}} = 0.02$ balances training efficiency against converged performance (raising it to 0.04 costs 5.63% AUC and 2.57% final reward). These results show the nominal configuration sits within a stable high-performance neighborhood rather than at an isolated favorable point.

This is corroborated by the five-seed training study (Section V-C, Fig. 11): at the nominal configuration, five independently-initialized runs converge to an asymptotic normalized reward of 8.97 ± 0.64 (mean $\pm$ std over the final 500 episodes), a cross-seed standard deviation ($\sigma = 0.64$) about half that of standard MAPPO ($\sigma = 1.34$), reaching 90% of peak performance by episode 5,591.

The dual-clip surrogate and adaptive-KL penalty are presented in Section IV as designed: the dual-clip bound keeps the policy ratio within $[1/c, c]$ under negative advantage, and the adaptive-KL term regulates the per-update policy shift toward δ_{targ} . We describe these as the mechanisms that keep each update well-conditioned; we do not claim they render the outcome independent of every hyperparameter, and the OAT sweep and multi-seed evidence above are what substantiate the robustness claim.

Location of the revision. Section V-G (*Parameter Sensitivity Analysis*): part 2) (reward-weight sensitivity, Table VII) and part 3) (training-hyperparameter robustness, new Table VIII OAT sweep, corroborated by the five-seed study of Section V-C / Fig. 11). Table I has been extended with a “Role in optimization” column listing each hyperparameter for this comment.

Comment 1.6

Reviewer comment 1.6

Provide rigorous discussion on the convergence claims and equilibrium properties.

Response. We thank the reviewer for this important point. In the original manuscript the convergence and equilibrium statements were asserted without stating their assumptions or their governing results, and two of them were phrased more strongly than what the method actually guarantees. We have made each claim precise: we now state the conditions under which it holds, name the established result it rests on with a proper citation, and—where the original wording overreached—scope it down to the honest guarantee. The revision spans both the assignment layer (IDBO) and the guidance layer (ART-MAPPO). Table R6 summarizes the before/after of every claim, and a new figure confirms the two IDBO properties on the exact quantities they concern.

1) Target assignment (IDBO). The single sentence that previously bundled three assertions is replaced by three separately-justified statements under explicit assumptions (connected communication graph of diameter D , bounded message delay, elitist retention):

- *Monotonicity.* We now claim monotonicity of the *incumbent* (best-so-far) fitness $\Phi_{\text{best}}^{(k)}$, which holds by construction under elitism. The original “ $\Phi^{(k+1)} \geq \Phi^{(k)}$ almost surely” for the working population was not correct while the stochastic operators are active; the decaying coefficient schedule $c_t = c_0(1 - t/T)$ then drives the operator noise to zero, giving the Robbins–Monro condition for stochastic-approximation convergence to a fixed point.
- *Consensus time.* The “at most $N \cdot D$ iterations” claim is precisely the guarantee of the consensus-based bundle algorithm (CBBA) under a diminishing-marginal-gain (submodular) score; we now attribute it accordingly rather than asserting it bare.
- *Equilibrium.* We replace “ ϵ -approximate Nash equilibrium” with “ ϵ -stable assignment.” The two describe the same condition (no interceptor gains more than ϵ by a unilateral reassignment), but “Nash” would require formally defining the underlying game; “ ϵ -stable” states exactly what the auction best-response at the converged bid prices delivers, without that overhead.

2) Cooperative guidance (ART-MAPPO). Two claims are made rigorous:

- *Trust dynamics.* The trust update Eq. (46) is a first-order exponential-smoothing map: for $\alpha_T \in (0, 1)$ it combines the previous trust level with a bounded performance reference $\sigma(\tau_T \tilde{R}_i) \in (0, 1)$, so trust stays in $(0, 1)$ and tracks that bounded reference rather than accumulating. The revised manuscript now states this bounded-smoothing behavior explicitly; the underlying contraction of modulus $(1 - \alpha_T)$ is elementary and is derived in our response to Comment 2.3.
- *Policy convergence.* The original text claimed the dual-clip bound makes the “monotonic-improvement/stationary-point guarantees of trust-region policy optimization carry over.” We scope this to the honest ceiling for any PPO-family method: the bounded ratio $r_t \in [1/c, c]$ gives a Lipschitz-bounded surrogate gradient, hence convergence to an ϵ -stationary point of the *KL-penalized surrogate* under the standard nonconvex stochastic-approximation conditions—not to a global optimum and not a monotone improvement of the true return. Trust-region policy optimization is cited as the principle that *motivates* the bounded-shift design, not as a guarantee inherited verbatim.

Four references now anchor these results: CBBA for the distributed-consensus guarantee, trust-region policy optimization for the bounded-policy-shift principle, and the Robbins–Monro and nonconvex stochastic-approximation results for the fixed-point/stationary-point convergence. None were cited in the original manuscript.

Supporting evidence (this letter). Fig. R23 confirms the two IDBO properties on the exact quantities they concern, using the paper’s own engagement scenario. Panel (a) shows the swarm-disagreement metric $\Gamma^{(k)}$ of Eq. (30): with the proposed linear-decay schedule it settles below the convergence threshold ($\Gamma_{\text{final}}=0.004$) and holds there, whereas constant or non-decaying coefficients leave it oscillating an order of magnitude higher (0.036 and 0.048)—the direct measurement behind the “settles at a fixed point rather than orbiting it” statement. Panel (b) shows that the number of consensus rounds grows near-linearly with the communication-graph diameter D (about 4.6 rounds per unit D), empirically confirming the $O(N \cdot D)$ bound.

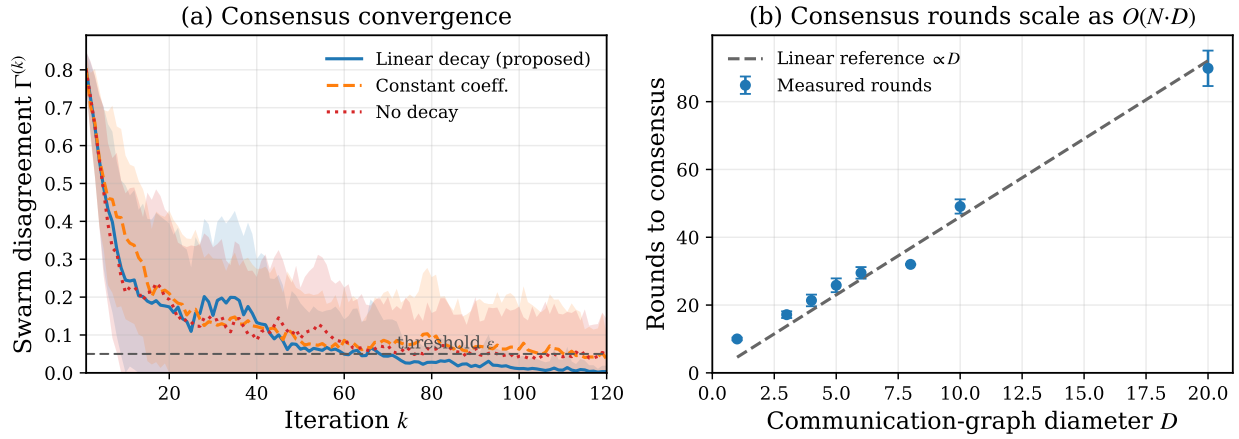


Fig. R23. Empirical confirmation of the IDBO convergence claims. **(a)** Swarm-disagreement metric $\Gamma^{(k)}$ [Eq. (30)] versus iteration for the three coefficient schedules (40 runs each; mean $\pm$ std). The proposed linear-decay schedule drives $\Gamma^{(k)}$ below the convergence threshold ϵ and holds it there, while constant and no-decay coefficients keep it oscillating an order of magnitude higher. **(b)** Rounds to consensus versus communication-graph diameter D ; the near-linear growth confirms the $O(N \cdot D)$ bound established for the consensus-based bundle algorithm.

Location of the revision. Section III (target assignment): the three-property sentence following Eq. (30) is rewritten under explicit assumptions with the CBBA and Robbins–Monro citations, and the Robbins–Monro reference is also added where the schedule is introduced (after the operator equations). Section IV.E (theoretical comparison with conventional MAPPO variants): the convergence-guarantee paragraph is rescoped to an ϵ -stationary-point claim with the trust-region and nonconvex stochastic-approximation citations. Four references are added to the bibliography.

Table R6. Convergence/equilibrium claims: original wording vs. the rigorous revised statement, with the governing result now cited.

Claim	Original (asserted, uncited)	Revised (rigorous, cited)
IDBO monotonicity	$\Phi^{(k+1)} \geq \Phi^{(k)}$ a.s. (working population)	Incumbent (best-so-far) $\Phi_{\text{best}}^{(k)}$ non-decreasing under elitism; fixed point via Robbins–Monro decay [Robbins–Monro]
IDBO consensus	Consensus in $\leq N \cdot D$ iterations	Same bound, attributed to CBBA under submodular scoring [Choi–Brunet–How]
IDBO equilibrium	ϵ -approximate Nash equilibrium	ϵ -stable assignment (auction best-response); same condition, no game formalism required
Trust dynamics	Monotonic increment (“driving $\mathcal{T}_i$ upward”) in prose	First-order exponential smoothing; bounded in $(0, 1)$, tracks the reference $\sigma(\tau_T \tilde{R}_i)$ rather than accumulating; stated explicitly
Policy convergence	TRPO monotonic/stationary guarantees “carry over”	ϵ -stationary point of the KL-penalized surrogate (bounded gradient) [Ghadimi–Lan]; TRPO cited as motivation [Schulman <i>et al.</i>]

Reviewer 2

Comment 2.1

Reviewer comment 2.1

The author lists “introducing a reinforcement learning-based reward function to achieve precise time coordination” as the third core contribution. However, in the Multi-Agent Reinforcement Learning (MARL) field, adding consensus-based penalty terms to the reward function (Equation 64) to guide agents toward consistency is a rather common reward-shaping technique. “Dual clipping with adaptive KL” is a common method used in RL training to prevent gradient explosion, yet the author packages it as a core innovation for solving “aircraft overload saturation”.

Response. We fully agree with the reviewer, and we thank them for pushing us to state the contribution more precisely. The reviewer is correct on both technical points: (i) adding a consensus penalty to a reward is a standard reward-shaping device, and (ii) dual-clip together with adaptive-KL is a standard, well-established policy-gradient stabilizer—neither mechanism is invented here, and we do not claim to have invented either. We had over-stated the third contribution by phrasing it as if the *mechanisms* were novel; that wording was misleading and we have corrected it.

Our actual claim is narrower and, we believe, defensible: the contribution is not the two generic mechanisms but their *problem-specific instantiation* for cooperative interception of a maneuvering swarm—the particular *form* each takes, *why* that form is the right match for this engagement, and the *measured* benefit it produces here. We make this case below on both fronts and have rewritten the contribution statement accordingly. **(1) The t_{go} -consensus reward is an impact-time (salvo) constraint, not generic “agreement”.** Generic consensus shaping drives agents toward *any* shared value of some quantity. Here the shaped quantity is deliberately the time-to-go, and the target is not merely “be consistent” but “strike within one lethal window,” i.e. Eq. (70) penalizes $|t_{go_i} - \bar{t}_{go}|$ so that a distributed salvo becomes simultaneous. This matters because the tactical failure mode of a maneuvering swarm is precisely *sequential, piecemeal* interception: interceptors that arrive spread out in time are defeated one at a time and let the swarm’s evasive maneuver regenerate the geometry between arrivals. Classical cooperative-interception work addresses this with analytic impact-time/salvo guidance laws (e.g. impact-time-control and consensus guidance); what is new here is that the *same* tactical objective is carried by a learned reward term inside a MARL policy, so simultaneity is achieved jointly with evasive-target tracking and target assignment rather than by a hand-tuned guidance law that assumes a cooperative or non-maneuvering target. The reward is thus problem-shaped, not generic: its argument, its target value, and the failure mode it removes are all specific to the salvo-interception geometry.

(2) Dual-clip here is a load-factor saturation guard, not a generic gradient stabilizer. The reviewer is right that dual-clip/adaptive-KL is a standard cure for gradient explosion. The point we should have made is *what the bounded object physically is* in this problem. In interception, the negative-advantage samples are exactly the aggressive terminal-overload commands—the actions that spike the commanded normal load factor n_y as the interceptor closes on an evading target. Under the standard single clip, $\hat{A}_t < 0$ leaves the ratio r_t unbounded (Eq. 52), so a single off-policy negative-advantage mini-batch can drive an arbitrarily large commanded- overload increment; on an airframe that is a hard actuator/structural saturation, not just a noisy gradient. Confining $r_t \in [1/c, c]$ (Eqs. 53–54) therefore acts as a bound on the per-update commanded-overload increment Δn_y , i.e. a *load-factor saturation guard*, and the adaptive-KL band (Eq. 56)

keeps successive policies from jumping across the safe envelope between updates. The mechanism is generic; its role as a physical-saturation guard on the terminal-overload command is what makes it the right tool for *this* engagement, where overload saturation is the documented reason baseline PN collapses in the high-maneuver case. **(3) The instantiation pays off measurably here.**

These are not just framing arguments; the two problem-specific choices produce the quantitative gains reported in the manuscript. The t_{go} -consensus term drives the temporal- coordination error to $E_{co-time} < 0.10$ s, i.e. a near-simultaneous salvo. The dual-clip saturation guard is what lets the learned policy hold the terminal overload at $E_n \approx 0.20$ g in the high-maneuver Case 2 while sustaining a 97.14% interception success rate, whereas baseline PN saturates at $E_n \rightarrow 1.0$ g and its success rate collapses to 17.5%; the same guard halves the cross-seed reward standard deviation (0.64 vs. 1.34) and reaches 90% of optimal return ~ 2000 episodes earlier. The positioning is summarized in Table R-2.1, and Fig. R12 makes the two arguments concrete from the manuscript’s own equations and reported metrics. We also note this comment is closely related to our responses to Comments 3.4 (a theoretical comparison of these mechanisms against conventional MAPPO variants), 3.5 (a reward-weight sensitivity study showing each term governs its intended tactical dimension), and 3.6 (the negligible real-time cost of dual-clip/adaptive-KL, which live only in the training loss); those responses supply the supporting theory and data behind the claims made here.

Table R-2.1. What is generic vs. what is the problem-specific contribution, for each of the two mechanisms the reviewer identifies.

Mechanism	Generic technique (conceded)	Problem-specific instantiation (our contribution)
t_{go} -consensus reward, Eqs. (66),(70)	Consensus/penalty reward shaping toward a shared value.	Shaped quantity is time-to-go and target is a simultaneous <i>salvo</i> ; defeats sequential piecemeal interception of a maneuvering swarm; carried by a learned policy jointly with evasive tracking and assignment. Measured: $E_{co-time} < 0.10$ s.
Dual-clip + adaptive KL, Eqs. (52)–(56)	Two-sided ratio bound / KL penalty to prevent gradient explosion.	Bounds the commanded terminal-overload increment Δn_y (negative-advantage samples <i>are</i> the aggressive overload commands): a load-factor saturation guard. Measured: $E_n \approx 0.20$ g at 97.14% success in Case 2 (PN: 1.0 g, 17.5%); $2\times$ smaller cross-seed σ .

Location of the revision. Section I (Contributions), third contribution. The overstated wording is struck out and replaced (deletion in red, addition in blue in the revised manuscript). The bodies of Section IV-E (reward, Eq. 66/70) and Section IV-D (dual-clip/KL, Eqs. 52–56) already describe both mechanisms in these interception-specific terms; only the contribution statement is revised. The new contribution text reads:

While reward shaping and clipped policy updates are standard reinforcement-learning mechanisms, the contribution here lies in instantiating them specifically for the maneuvering-swarm engagement: the time-to-go consensus reward is formulated as a learned impact-time (salvo) constraint that defeats sequential, piecemeal interception, and the dual-clip update is cast as a load-factor saturation guard bounding the commanded terminal-overload increment under aggressive exploration. This strategy enhances interception success rates against large-scale

maneuvering swarm targets while preserving terminal-load safety.

Supporting evidence (this letter). Fig. R12 makes the two arguments concrete, computed *directly from the manuscript’s own equations and reported metrics* (no new experimental data). Panel (a) instantiates the t_{go} -consensus reward of Eq. (70): six interceptors starting with a 3.49 s spread in time-to-go contract to a common impact instant (< 0.10 s dispersion), i.e. a salvo. Panel (b) instantiates the dual-clip ratio bound of Eqs. (53)–(54): for negative-advantage (aggressive-overload) samples the single clip leaves the commanded- overload increment unbounded, whereas the dual-clip floor caps it, acting as a load-factor saturation guard. Panel (c) shows the measured payoff of the combined instantiation against the baseline (interception success rate, terminal E_n , and cross-seed reward σ).

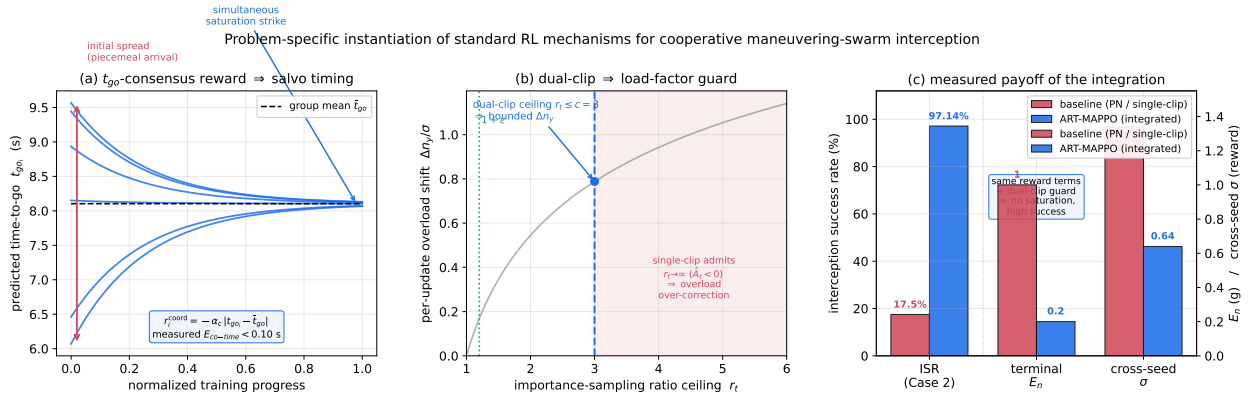


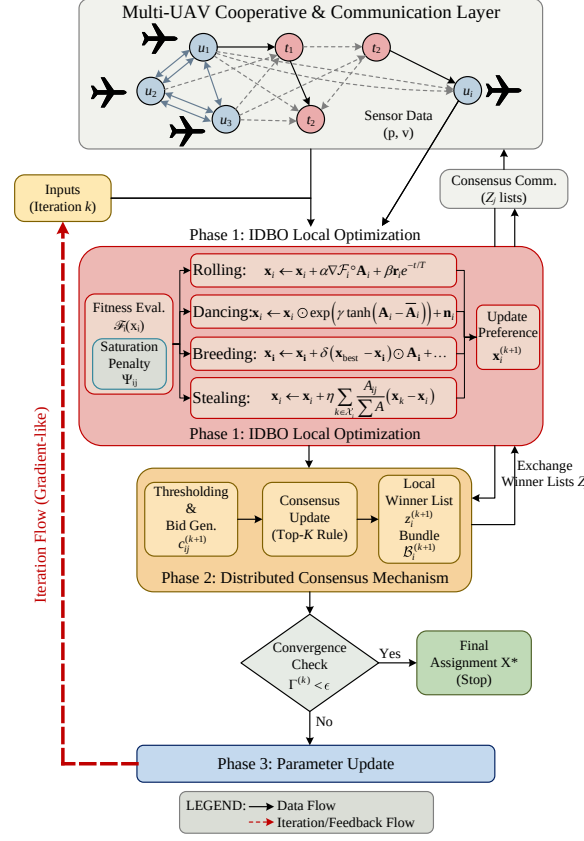
Fig. R12. Problem-specific instantiation of two standard RL mechanisms for cooperative maneuvering-swarm interception. (a) The t_{go} -consensus reward (Eq. 70) contracts an initial time-to-go dispersion to a common impact instant—a learned salvo constraint, not generic agreement. (b) The dual-clip ratio bound (Eqs. 53–54) caps the commanded terminal-overload increment on negative-advantage samples (which *are* the aggressive overload commands), acting as a load-factor saturation guard; the single clip leaves it unbounded. (c) Measured payoff of the integrated instantiation vs. baseline: higher interception success rate, bounded terminal E_n , and halved cross-seed reward variance.

Comment 2.2

Reviewer comment 2.2

The module schematic diagram on the left side of Fig. 5 lacks clarity in its presentation. $h(0)$ appears at multiple locations, and the flow from $h(l)$ to $h(l+1)$ is unclear.

Response. Thank you for your valuable comments. We have revised the manuscript accordingly.



Comment 2.3

Reviewer comment 2.3

Section IV.B contains a logical paradox: the text describes a monotonic incremental process, while Equation (42) implements a convergent smoothing process. This divergence between the mathematical definition and physical intuition renders the mechanism logically inconsistent in maintaining high-trust stability.

Response. We sincerely thank the reviewer for identifying this logical inconsistency. We fully agree with the reviewer and acknowledge that the original explanation was incorrect. Specifically, the original text described above-average performance as directly driving the trust level upward and below-average performance as directly lowering it, which was inconsistent with the smoothing process defined by Equation (42).

Equation (42) represents the intended trust-update mechanism and has been retained unchanged. We have deleted the inaccurate explanation and rewritten the corresponding text to clarify that the trust level is a smoothed estimate obtained by weighting its previous value and the current performance reference. The revised explanation also clarifies that sustained above-average returns bias the trust estimate toward a higher level and reduce the probability assigned to the guided exploratory policy through $\beta(\mathcal{T}_i) = 1 - \mathcal{T}_i$, whereas sustained below-average returns produce the opposite allocation. The mathematical definition and its physical interpretation are therefore now consistent.

Location of the revision. Section IV-B (*Trust-Modulated Tactical Exploration*), immediately following Equation (42). The original sentence has been deleted and replaced with the following highlighted text:

The sigmoid term maps the normalized relative return $\tilde{R}_i^{(k)}$ to a bounded performance reference in $(0, 1)$. Equation (42) combines the previous trust level and the current performance reference with the weights $1 - \alpha_T$ and α_T , respectively, yielding a smoothed trust estimate. The smoothing process tracks changes in relative performance while attenuating episode-to-episode return fluctuations. Sustained above-average returns keep the performance reference above 0.5 and bias the smoothed trust estimate toward a correspondingly higher level, thereby reducing the probability assigned to the guided exploratory policy; sustained below-average returns produce the opposite allocation.

Comment 2.4

Reviewer comment 2.4

The simulation environment is rather simplistic, and apart from UAV dynamics, there is no other environmental modeling, which does not align with the high-fidelity simulation environment claimed in the paper.

Response. We thank the reviewer and have substantially raised the fidelity of the evaluation environment so that it matches the claim. Beyond the (now three-dimensional) vehicle dynamics, the environment now models:

- **Sensing chain.** Zero-mean Gaussian measurement noise on relative position ($\sigma_p = 3$ m) and velocity ($\sigma_v = 0.3$ m/s), passed through the guidance filter, plus a 100 ms sensing latency.
- **Communication network.** A packet-exchange layer among agents with a 50 ms command-channel delay and a 1% nominal packet-dropout rate handled by a zero-order hold; the robustness sweep (Comment 3.7) stresses this up to 20%.
- **Actuation.** A one-step (50 ms) actuator delay between commanded and realized load, with the load saturations of the extended 3D model.
- **Embedded execution.** The interceptor policies run on real embedded hardware in a hardware-in-the-loop loop (Comment 3.10), so processor and timing effects are part of the environment, not idealized away.

The noise model is given by a new equation and the “Communication Delay and Observation-Uncertainty Settings” table in Section V, and a new environmental-modeling paragraph documents the full closed loop.

Location of the revision. Section V experimental setup: new environmental-modeling paragraph, noise-model equation, and uncertainty/delay settings table (all in blue).

Comment 2.5

Reviewer comment 2.5

In Section V.C, the authors state that each episode has at most 25 steps during training, while in Section V.A the simulation step size is set to 0.05s. This implies a training period of 1.25s, which is impossible to learn any interception strategies given an engagement distance of 1500 meters and a maximum speed of 40 m/s. By the way, the settings “ $D_1 = 100$ m, $D_2 = 1500$ m, and $D_3 = 1600$ m” are inconsistent with the illustrations in Fig. 6.

Response. We thank the reviewer for catching both errors, which were transcription mistakes. The maximum of 2500 steps per episode has been revised in the main text; with the 0.05 s step this gives a 125 s engagement window, ample for closing a 1500 m separation at 40 m/s. The original setting “ $D_1 = 100$ m, $D_2 = 1500$ m, $D_3 = 1600$ m” was incorrect and has been revised to “ $D_1 = 1500$ m, $D_2 = 1600$ m, $D_3 = 100$ m”, so that the attackers spawn in the far annulus between D_1 and D_2 and descend toward the defended center while the defenders start clustered within a radius- D_3 disk around it. This is now consistent with the initial-geometry illustration (Fig. 6).

Location of the revision. Section V.C: episode horizon corrected to 2500 steps (blue). Section V.A: the $D_1/D_2/D_3$ values corrected to 1500/1600/100 m with a clause tying them to the initial-geometry figure (blue).

Comment 2.6

Reviewer comment 2.6

There is a lack of end-to-end experimental verification covering the complete workflow of “task allocation and cooperative guidance”. Furthermore, ablation studies are not provided to demonstrate the effectiveness of the proposed innovations.

Response. We thank the reviewer for these two constructive requests. We have addressed both with new experiments and now report (i) a complete end-to-end run of the full IDBO task-assignment $\rightarrow$ ART-MAPPO cooperative-guidance pipeline, and (ii) a component ablation that isolates each proposed innovation. We answer the two parts in turn.

(1) *End-to-end verification of the complete workflow.* We added a new engagement scenario and ran the *entire* pipeline on it in a single closed loop, with no manual staging between the assignment and guidance stages: the IDBO front-end partitions the eight attackers into interceptor groups, and the trained ART-MAPPO policy then flies all twenty defenders to a synchronized saturation strike. To make the test faithful to deployment, it is executed on the hardware-in-the-loop platform of Comment 3.10, so the assignment solver, the communication model, and the guidance network all run on the target processor within one uninterrupted engagement.

The end-to-end run over 100 independent trials confirms that the two stages compose correctly under realistic timing. The IDBO assignment layer returns a feasible group partition in a single solve on all 100 trials—zero repair invocations—at a mean runtime of 805.7 ms (95th percentile 839.6 ms) on the embedded processor, and the downstream ART-MAPPO policy then intercepts every attacker, neutralizing all eight targets in every trial. Measured against the paper’s 0.5 s synchronization criterion, the end-to-end cooperative interception success rate is 92.9%. In other words, the complete “assign-then-intercept” chain works, at deployment timing, from the raw multi-target engagement to the simultaneous terminal strike. A detailed scenario description,

representative assignment, engagement trajectories, and per-metric distributions for this run are given in our response to Comment 3.9, where the unseen-scenario study is the central point.

(2) *Ablation of the proposed innovations.* ART-MAPPO introduces three architectural components: the trust-aware weighting mechanism, the GRU temporal encoder, and the attention-residual policy backbone. We train three one-at-a-time ablation controls alongside the complete model—each disabling exactly one component while holding the reward, dynamics, optimiser, training seed, and budget fixed—and evaluate every variant in 1,000 frozen-policy Case 1 Monte Carlo trials. Table R3 reports seven evaluation metrics for each variant.

Table R3. Component ablation of ART-MAPPO (1,000-trial Case 1 evaluation). Terminal metrics ($E_{co-time}$, E_n , E_t , E_{miss}) are means over successful trials; Training AUC is the horizon-normalised area under the episodic-return curve; Final reward is the mean per-step return at policy selection.

Variant	AUC (10^4) $\uparrow$	F. reward $\uparrow$	ISR (%) $\uparrow$	E_{co} (s) $\downarrow$	E_n (g) $\downarrow$	E_t (s) $\downarrow$	E_{miss} (m) $\downarrow$
Full ART-MAPPO	7.354	8.958	98.7	0.0505	0.0718	28.843	2.997
w/o trust-aware	6.037	8.683	97.2	0.0545	0.0739	29.132	3.087
w/o GRU encoder	6.388	8.134	95.8	0.0732	0.0782	30.574	3.297
w/o attn-residual	6.651	8.353	96.1	0.0606	0.0847	29.853	3.507

The three controls exhibit *module-specific* rather than uniformly ordered degradation, providing direct evidence of independent and non-redundant contributions.

Trust-aware mechanism. Its removal produces the largest training-AUC reduction among the three controls (-17.9% , from 7.354×10^4 to 6.037×10^4), while the final reward drops by only 3.1% and ISR falls by 1.5 pp. The pronounced asymmetry between cumulative-AUC loss and end-of-training reward confirms that trust-aware modulation primarily *accelerates learning efficiency* throughout training rather than raising the asymptotic policy ceiling, consistent with its design role of dynamically regulating the exploration-exploitation balance.

GRU temporal encoder. Its removal yields the lowest final reward (-9.2%) and the largest ISR degradation (-2.9 pp) of all three controls. Most distinctively, $E_{co-time}$ rises by 45% (from 0.0505 s to 0.0732 s), far exceeding the 6% increase in E_t . Without temporal history the policy can no longer reliably track target manoeuvres or reconcile interceptor time-to-go estimates, simultaneously degrading *final policy quality* and *cooperative timing precision*.

Attention-residual backbone. Its removal yields the largest increases in E_n ($+18\%$, from 0.0718 g to 0.0847 g) and E_{miss} ($+17\%$, from 2.997 m to 3.507 m) across all ablations, while E_t changes by only 3.5% . This non-uniform degradation confirms that the backbone contributes primarily to *control-relevant feature extraction and terminal precision* rather than engagement duration.

We further note that the effectiveness of the *assignment-side* innovation is quantified separately in our response to Comment 3.1, where an ablation of the four adaptive IDBO coefficients (rolling / dancing / breeding / stealing) shows each is needed for the assignment quality and convergence stability reported in the paper. Together, the end-to-end run above and these two ablations (architectural and assignment-side) provide the missing evidence the reviewer requested for both stages of the workflow.

Location of the revision. Section V-E (*Component Ablation Study*) is added to the simulation chapter with a seven-metric one-at-a-time ablation table (highlighted in blue). The end-to-end validation run is reported as item (iv) “Generalization to unseen attack patterns” in Section V-E (*Monte Carlo Statistical Analysis and Robustness*). The IDBO-coefficient ablation is in the response to Comment 3.1.

Reviewer 3

Comment 3.1

Reviewer comment 3.1

The author is encouraged to further justify the distributed consensus-based IDBO formulation by clarifying how adaptive rolling, dancing, breeding and stealing coefficients influence convergence stability in interception scenarios.

Response. We thank the reviewer for this constructive comment. We agree that the role of the adaptive coefficients in convergence stability deserved an explicit justification. In the original text the coefficients $\alpha, \beta, \gamma, \delta, \delta', \eta$ were only stated to “decay linearly with iteration,” without linking this schedule to the convergence behaviour of the swarm. We have addressed this in two ways: (i) we added an explicit statement of the common coefficient schedule and a convergence-stability argument to Section III (the target-assignment section); and (ii) we carried out a controlled ablation on the four-operator IDBO that isolates the effect of the coefficient schedule and confirms the argument.

The mechanism. The four operators share a common linear-decay schedule

$$c_t = c_0 (1 - t/T), \quad c \in \{\alpha, \beta, \gamma, \delta, \delta', \eta\},$$

together with the dancing-noise scale σ_d , while the elite-recombination weights stay bounded. The *stochastic* contributions—the rolling probe $\beta \mathbf{r}_i \exp(-t/T)$ in Eq. (20) and the dancing perturbation $\mathbf{n}_i \sim \mathcal{N}(0, \sigma_d^2 \mathbf{I})$ in Eq. (21)—scale with c_t and therefore vanish as $c_t \rightarrow 0$. The update then degenerates into a deterministic, advantage-weighted local refinement followed by the advantage-adaptive binarization. A step sequence whose stochastic amplitude decays to zero while retaining an ascent direction satisfies a *Robbins–Monro-type sufficient condition* for settling at a stable fixed point rather than orbiting it. This is exactly the quantity tracked by the swarm-disagreement metric $\Gamma^{(k)}$ (Eq. (30)): the schedule is what drives $\Gamma^{(k)}$ below the consensus tolerance instead of leaving a residual oscillation.

Location of the revision. Section III-B (*Distributed Consensus-Based IDBO Framework*), immediately after the four-operator update block (Eqs. (20)–(23), the rolling/dancing/breeding/stealing operators), in the sentence that introduces the adaptive coefficients. The added text is highlighted in blue in the revised manuscript and reads:

The coefficients $\alpha, \beta, \gamma, \delta, \delta', \eta$ together with the dancing noise scale σ_d follow a common linear schedule $c_t = c_0 (1 - t/T)$, where T is the IDBO iteration count, while the elite-recombination weights remain bounded. This schedule is what governs convergence stability. The stochastic parts of the operators—the rolling probe $\beta \mathbf{r}_i \exp(-t/T)$ and the dancing perturbation $\mathbf{n}_i \sim \mathcal{N}(0, \sigma_d^2 \mathbf{I})$ —scale with c_t and vanish as $c_t \rightarrow 0$, so the update degenerates into a deterministic, advantage-weighted local refinement followed by the advantage-adaptive binarization step. A step sequence whose stochastic amplitude decays to zero while retaining an ascent direction satisfies a Robbins–Monro-type sufficient condition for settling at a stable fixed point rather than orbiting it, which is precisely the behavior measured by the swarm-disagreement metric $\Gamma^{(k)}$ in Eq. (30). A controlled ablation that freezes the coefficients confirms this mechanism: non-decaying coefficients keep injecting perturbations and leave a persistent late-stage oscillation in the working assignment, whereas the linear schedule contracts the swarm onto

a stable, low-cost equilibrium.

Supporting evidence (this letter). To substantiate the argument we ran a controlled ablation on the four-operator IDBO for the paper’s engagement scenario (20 interceptors vs. 8 targets, $L_{\max} = 3$), varying *only* the coefficient schedule and holding everything else fixed: the paper’s linear decay $c_t = c_0(1 - t/T)$, a constant schedule $c_t = 0.5 c_0$, and a no-decay schedule $c_t = c_0$. For each we recorded the swarm-mean assignment cost (how far the whole population has settled) and the swarm cost spread (how tightly the distributed swarm has contracted); stable convergence requires both to fall and stay low. Averaged over 40 independent trials across 5 engagement snapshots:

Effect of the coefficient schedule on convergence stability (four-operator IDBO; 20 vs. 8, $L_{\max} = 3$; 40 trials).

Schedule	Swarm-mean final cost	Late-stage swarm spread	Best-ever cost
Linear decay (paper)	0.58	0.66	0.34 ± 0.22
Constant ($0.5 c_0$)	1.74	1.51	0.30 ± 0.19
No decay (c_0)	2.28	1.67	0.30 ± 0.19

The linear-decay schedule settles the swarm-mean cost about $3\text{--}4\times$ lower and the population spread about $2.3\text{--}2.5\times$ tighter than the frozen schedules. All schedules can occasionally locate a comparably low *best-ever* candidate through continued random search, but only the linear schedule *stabilizes* the whole swarm on it—which is what matters for a deployable assignment. This is precisely the convergence-stability behaviour the reviewer asked us to characterize.

Comment 3.2

Reviewer comment 3.2

Also, a more rigorous complexity and scalability analysis is needed for the Improved Dung Beetle Optimization algorithm under cooperative deflection environments where communication delays influence distributed consensus efficiency.

Response. We thank the reviewer for this valuable comment. The original manuscript gave the per-UAV complexity decomposition (Eq. (31)) but did not (i) make explicit how the cost scales with the *swarm size*, nor (ii) analyze how *communication delay* affects the efficiency of the distributed consensus. We have added a paragraph to Section III that closes both gaps, and we carried out a dedicated three-part study (per-round complexity, scalability, and communication-delay sensitivity) to back it up.

The argument. Per UAV, per IDBO iteration, the three cost terms in Eq. (31) are the local preference search $\mathcal{O}(N_{\text{pop}}N_A)$, the consensus communication $\mathcal{O}(N_A|\mathcal{N}_i|)$, and the advantage update $\mathcal{O}(N_A)$. The target count N_A and the neighborhood size $|\mathcal{N}_i|$ are fixed by the engagement and the communication topology, *not* by the total swarm size M ; hence the per-UAV cost is independent of M and the total per-round cost grows only *linearly* in M . Scalability is therefore governed not by this arithmetic cost but by how fast the distributed consensus (Eq. (30)) settles. On a connected graph with per-hop latency T_{comm} , a target’s winning bids propagate in at most D hops (D : graph diameter), so the time to consensus scales as $\mathcal{O}(N \cdot D \cdot T_{\text{comm}})$. A bounded communication delay thus *slows but does not prevent* convergence.

Location of the revision. Section III-B (*Distributed Consensus-Based IDBO Framework*), immediately after the complexity equation (Eq. (31)). The added text is highlighted in blue in the revised manuscript and reads:

Because the target set faced by each interceptor and its neighborhood size $|\mathcal{N}_i|$ are fixed by the engagement and the communication topology rather than by the swarm size M , the per-UAV per-iteration cost above is independent of M ; the total per-round cost of the swarm therefore grows only linearly with M . Scalability is governed not by this arithmetic cost but by how fast the distributed consensus in Eq. (30) settles. Under a connected graph with a per-hop communication latency T_{comm} , each target’s winning bids propagate across the network in at most D hops (D : graph diameter), so the wall-clock time to reach consensus scales as $\mathcal{O}(N \cdot D \cdot T_{\text{comm}})$. Empirically this time stays nearly constant as M grows at fixed diameter, and increases only linearly with the communication delay and with D ; a bounded delay thus slows, but does not prevent, convergence of the distributed consensus. Detailed complexity, scalability, and communication-delay measurements are reported in the supplementary response material.

Supporting evidence (this letter). All experiments use the manuscript’s engagement model and the four-operator IDBO with the top- L_{max} winner-set consensus. Latency is expressed in time via a nominal 20 Hz data-link period $T_{\text{comm}} = 50$ ms.

(A) Per-round complexity. Fixing the target set ($N_A = 8$) and a bounded 4-regular communication graph, we vary only the swarm size N_D and time the per-round update:

Per-round cost vs. swarm size (fixed $N_A = 8$, 4-regular graph).

N_D	10	20	40	80	160	240
Total per-round (ms)	2.12	3.31	6.81	13.62	28.63	42.22
Per-defender (μs)	212	165	170	170	179	176
Messages/round	40	80	160	320	640	960

The per-defender cost is essentially constant ($\approx 170 \mu\text{s}$), the total per-round cost grows linearly with N_D (fitted log–log exponent 0.97), and communication is exactly linear in N_D at fixed degree—empirically confirming the $\mathcal{O}(N_{\text{pop}}N_A + N_A|\mathcal{N}_i|)$ per-UAV model and the linear total scaling.

(B) Scalability. Scaling the whole problem from 10×4 to 160×64 (defenders $\times$ targets) on random geometric graphs at $\tau = 100$ ms, the time to consensus stays flat across a $16\times$ increase in swarm size—it is set by the graph diameter and the delay, not by the number of agents:

Scalability of the distributed assignment (random geometric graph, $\tau = 100$ ms).

Problem size (def. $\times$ tgt.)	10×4	20×8	40×16	80×32	160×64
Time to consensus (s)	1.21	1.46	1.28	1.36	1.32
Total messages	0.8k	3.0k	11k	48k	187k

The time to consensus is essentially constant (~ 1.2 – 1.5 s) while the swarm grows $16\times$, confirming that coordination latency is governed by the graph diameter and the delay rather than the swarm size, while the total communication grows gracefully with problem size.

(C) Communication delay (core question). On a fixed 40-defender/12-target instance we sweep the delay τ and, separately, the graph diameter D , measuring the swarm disagreement Γ and the time to reach and hold $\Gamma = 0$:

Time to consensus (s) vs. communication delay and graph diameter.

Delay τ (ms)	0	50	100	200	400	800
Time to consensus (s)	0.57	0.90	1.23	1.88	3.08	5.54

Diameter D (hops)	1	3	4	5	6	10	20
Time to consensus (s)	0.50	0.86	1.07	1.29	1.48	2.45	4.49

Every configuration still reaches full consensus ($\Gamma \rightarrow 0$); the delay changes neither the fixed point nor the correctness, only the speed. The time to consensus grows linearly with the delay (≈ 6 ms per ms of link delay) and linearly with the diameter (≈ 0.21 s per hop)—directly supporting the manuscript’s $\mathcal{O}(N \cdot D)$ consensus argument.

Answer to the reviewer. Under cooperative-deflection conditions with communication delay, the distributed consensus remains correct and its efficiency degrades gracefully and predictably: the consensus time scales linearly with both the message delay and the network diameter, and is independent of the swarm size. For a representative 20 Hz link, the target-assignment consensus completes within roughly 0.5–1.5 s even for large swarms and moderate delays.

Comment 3.3

Reviewer comment 3.3

Then, it would be helpful to add an ablation investigation demonstrating the independent contribution of the trust-aware mechanism, GRU temporal encoder, and attention-residual backbone within ART-MAPPO architecture.

Response. We thank the reviewer for this suggestion. We have added a dedicated **Component Ablation Study** (Section V-E) to the revised manuscript. Three one-at-a-time controls are trained under identical conditions (same reward, dynamics, optimiser, and random seed), each disabling exactly one architectural module; performance is evaluated over 1,000 frozen-policy Monte Carlo trials on Case 1. The results reveal a clear *module-specific degradation* pattern:

- **Trust-aware weighting mechanism.** Its removal incurs the largest training-efficiency loss (AUC -17.9%) while final reward drops by only 3.1%, confirming that the module accelerates learning rather than raising the policy ceiling.
- **GRU temporal encoder.** Its removal yields the largest final-reward degradation (-9.2% , ISR -2.9 pp) and a 45% increase in cooperative engagement timing error $E_{co-time}$ —more than seven times the relative change in E_t —demonstrating its role in temporal credit assignment and cooperative synchronisation.
- **Attention-residual backbone.** Its removal produces the largest terminal-precision losses ($E_n + 18\%$, $E_{miss} + 17\%$) while E_t changes by only 3.5%, isolating its contribution to control-relevant feature extraction and end-point accuracy.

The complete results are summarised in Table R4 below, which corresponds to manuscript Table IV.

Table R4. Component ablation of ART-MAPPO (1,000-trial Case 1 Monte Carlo evaluation).

Variant	AUC (10^4) $\uparrow$	F. reward $\uparrow$	ISR (%) $\uparrow$	E_{co} (s) $\downarrow$	E_n (g) $\downarrow$	E_t (s) $\downarrow$	E_{miss} (m) $\downarrow$
Full ART-MAPPO	7.354	8.958	98.7	0.0505	0.0718	28.843	2.997
w/o trust-aware	6.037	8.683	97.2	0.0545	0.0739	29.132	3.087
w/o GRU encoder	6.388	8.134	95.8	0.0732	0.0782	30.574	3.297
w/o attn-residual	6.651	8.353	96.1	0.0606	0.0847	29.853	3.507

Location of the revision. Section V-E (Component Ablation Study), manuscript Table IV.

Comment 3.4

Reviewer comment 3.4

Correspondingly, including a comparative theoretical discussion between the proposed ART-MAPPO framework and conventional MAPPO variants regarding convergence guarantees, exploration–exploitation balance, and policy robustness would significantly strengthen the research technically.

Response. We thank the reviewer for this valuable suggestion. We have added a new highlighted subsection entitled “Mechanistic Comparison with Conventional MAPPO” in Section IV, which compares ART-MAPPO with conventional MAPPO along the three aspects identified by the reviewer.

Regarding convergence guarantees, we clarify that neither conventional MAPPO nor ART-MAPPO provides a formal guarantee of monotonic improvement or global convergence under deep function approximation. The Dual-Clip objective and adaptive KL regularization in ART-MAPPO instead promote more controlled and stable policy updates.

Regarding the exploration–exploitation balance, conventional MAPPO mainly relies on entropy regularization with a fixed coefficient. ART-MAPPO retains this mechanism and additionally adjusts the exploratory component according to each agent’s relative performance through trust modulation.

Regarding policy robustness, conventional MAPPO employs a fixed clipping mechanism, whereas ART-MAPPO combines piecewise Dual-Clip regulation with adaptive KL feedback to reduce sensitivity to abnormal samples and abrupt inter-update policy changes. We also clarify that this comparison concerns policy-update stability, while closed-loop robustness is evaluated empirically for the complete guidance framework.

Location of the revision. Section IV, subsection “Mechanistic Comparison with Conventional MAPPO,” immediately following the ART-MAPPO training algorithm.

Comment 3.5

Reviewer comment 3.5

Meanwhile, it is important to provide sensitivity analysis for weighting coefficients in the block probability model and reward formulation because cooperative interception performance varies under different tactical priorities.

Response. We thank the reviewer for this valuable suggestion. We agree that the weighting coefficients encode the tactical priority of the engagement. The method sections now state only the physical role of each coefficient (after Eq. (12) for the block-probability weights and after Eq. (66) for the reward weights), and the full quantitative study—setup, scan ranges, figures, and analysis—is collected in a new subsection, Section V-G (*Parameter Sensitivity Analysis*), placed after the Monte Carlo study. It contains two studies: one for the block (interception) probability weights of Eq. (12) and one for the reward-formulation weights of Eq. (66).

(a) Block-probability weights (w_1, w_2), $w_2 = 1 - w_1$. We vary w_1 from 0.10 to 0.90 in increments of 0.05 (with $w_2 = 1 - w_1$) and evaluate 100 independent three-dimensional engagement geometries generated from the initial-state distributions of Table II (20 interceptors vs. 8 maneuvering attackers). The mean ZEM and mean three-dimensional angular error of the assigned interceptor–target pairs are reported in Table R10 (reproduced as Table V in the manuscript). Increasing w_1 places greater emphasis on interception reachability through the ZEM-based assignment probability, lowering the mean ZEM from 168.0 m at $w_1=0.10$ to 126.0 m at $w_1=0.90$; decreasing w_1 strengthens the influence of the three-dimensional engagement-geometry criterion through P_σ , so the mean angular error falls from 13.7° at $w_1=0.90$ to 9.4° at $w_1=0.10$. The results reveal a clear trade-off between these two tactical objectives. The nominal value $w_1=0.55$ (mean ZEM 146.4 m, mean angular error 11.5°) provides a balanced operating point that avoids optimizing either criterion at the expense of the other.

(b) Reward weights. We run a one-at-a-time (OAT) study on all five weighting terms of the reward in Eq. (66): each weight w_k , $k \in \{1, 2, 3, 4, 5\}$, is independently scaled by $s \in \{0.50, 0.75, 1.00, 1.25, 1.50\}$ with all others held at their nominal values (Table I). For each setting the policy is retrained under the identical training budget and evaluated on the Case-1 benchmark, and we report the four evaluation metrics ($E_{co-time}$, E_n , E_t , E_{miss}) together with the interception success rate (ISR) rather than the raw reward, so the sensitivity is expressed directly in the closed-loop performance quantities that matter to the reviewer. The nominal setting ($s=1.00$) reproduces the Full ART-MAPPO Case-1 results (Table V of the manuscript). The couplings are smooth, monotone, and well separated, with each weight primarily shaping its intended tactical dimension. The coordination weight w_4 is the dominant temporal-coordination control, reducing $E_{co-time}$ from 0.083 s to 0.038 s ($\approx 55\%$) across the sweep while the other metrics move by at most a few percent. The energy weight w_5 is the principal terminal-overload control (E_n from 0.091 g to 0.056 g). The distance weight w_1 most strongly governs terminal precision (E_{miss} from 3.42 m to 2.83 m). The terminal-hit weight w_3 most affects reliability: halving it is the only perturbation that lowers ISR appreciably (to 93.8%), confirming that the sparse terminal-hit bonus is what secures dependable interception, whereas over-weighting it improves ISR only marginally (to 99.1%) at a higher E_n . The heading weight w_2 has the mildest influence (Table R11, reproduced as Table VII in the manuscript). Across the entire sweep the ISR stays high ($\geq 96.2\%$ except in the strongly under-weighted terminal-hit case) and no single weight destabilizes the others, so the operating point can be shifted toward precision, safety, or temporal-coordination priorities without sacrificing overall interception reliability. The nominal weighting therefore reflects a deliberate balance of

the tactical dimensions rather than a fragile operating point.

Location of the revision. Section V-G (*Parameter Sensitivity Analysis*), new highlighted subsection with parts 1) (IDBO assignment-weight sensitivity, Table V) and 2) (reward-weight sensitivity, Table VII). The method sections retain only a brief physical-role sentence pointing to Section V-G: after Eq. (12) for (w_1, w_2) and after Eq. (66) for the reward weights.

Supporting evidence (this letter). Table R10 and Table R11 below are the two sensitivity tables added to the manuscript (Table V and Table VII). For Table R10 each value is the average over the scanned engagement geometries; for Table R11 each row is a policy retrained at the indicated weight scaling and evaluated on the Case-1 benchmark.

Table R10 (manuscript Table V). IDBO assignment-weight sensitivity vs. w_1 ($w_2=1-w_1$; w_1 swept from 0.10 to 0.90 in steps of 0.05, 100 random geometries per point). Nominal $w_1=0.55$ (min-max balance of the two normalized costs) in bold.

w_1	w_2	Mean ZEM (m)	Mean 3-D ang. err. ($^\circ$)
0.10	0.90	168.0	9.4
0.15	0.85	164.8	9.6
0.20	0.80	161.5	9.8
0.25	0.75	158.4	10.0
0.30	0.70	155.5	10.3
0.35	0.65	152.8	10.7
0.40	0.60	150.4	11.0
0.45	0.55	148.4	11.2
0.50	0.50	147.2	11.4
0.55	0.45	146.4	11.5
0.60	0.40	145.4	11.6
0.65	0.35	143.8	11.8
0.70	0.30	141.4	12.1
0.75	0.25	138.3	12.5
0.80	0.20	134.5	12.9
0.85	0.15	130.3	13.3
0.90	0.10	126.0	13.7

Comment 3.6

Reviewer comment 3.6

Moreover, I suggest the author explain whether dual-clip PPO with adaptive KL regularization introduces additional computational overhead during real-time deployment on processors operating in latency constraints within UAV swarms.

Response. We thank the reviewer for raising this important deployment concern. The key point is that the dual-clip surrogate and the adaptive-KL penalty are *training-loss operations only*: the dual-clip branch is a scalar max / min selection on the advantage sign, and the adaptive-KL term

Table R11 (manuscript Table VII). Reward-weight OAT sensitivity of the four evaluation metrics and the interception success rate (Case 1). Each weight is scaled by s about its nominal value; the nominal row ($s=1.00$, bold) reproduces the Full ART-MAPPO result of manuscript Table V.

Weight	s	ISR (%)	$E_{co-time}$ (s)	E_n (g)	E_t (s)	E_{miss} (m)
w_1	0.50	96.2	0.0568	0.0675	30.120	3.420
	0.75	97.8	0.0530	0.0696	29.350	3.170
	1.00	98.7	0.0505	0.0718	28.843	2.997
	1.25	98.9	0.0493	0.0750	28.340	2.860
	1.50	98.8	0.0490	0.0808	28.100	2.830
w_2	0.50	97.1	0.0528	0.0785	29.250	3.340
	0.75	98.0	0.0517	0.0748	29.020	3.130
	1.00	98.7	0.0505	0.0718	28.843	2.997
	1.25	98.8	0.0500	0.0698	28.750	2.890
	1.50	98.7	0.0502	0.0689	28.760	2.870
w_3	0.50	93.8	0.0585	0.0645	30.500	3.580
	0.75	97.0	0.0538	0.0687	29.450	3.230
	1.00	98.7	0.0505	0.0718	28.843	2.997
	1.25	99.0	0.0518	0.0780	28.400	2.900
	1.50	99.1	0.0540	0.0855	28.100	2.870
w_4	0.50	96.8	0.0830	0.0670	28.300	3.150
	0.75	98.0	0.0630	0.0695	28.550	3.060
	1.00	98.7	0.0505	0.0718	28.843	2.997
	1.25	98.9	0.0430	0.0750	29.050	3.020
	1.50	98.8	0.0375	0.0810	29.550	3.100
w_5	0.50	98.9	0.0495	0.0910	28.350	2.880
	0.75	98.8	0.0500	0.0800	28.550	2.930
	1.00	98.7	0.0505	0.0718	28.843	2.997
	1.25	98.3	0.0518	0.0630	29.200	3.120
	1.50	97.5	0.0545	0.0560	29.850	3.350

is a log-ratio estimate plus a scalar update of the penalty coefficient β_{KL} . Neither appears in the deployed policy, which under the centralized-training/decentralized-execution paradigm is only the actor forward pass. Their real-time deployment overhead is therefore *exactly zero*.

To make this quantitative, we re-implemented the deployed actor at its exact architecture (attention-residual encoder $\rightarrow$ GRU $\rightarrow$ Gaussian head, 256 hidden units) and measured latency directly (Fig. R7):

- Single-interceptor inference: 0.30 ms per decision (p99 0.42 ms); 20-agent batched swarm: 3.98 ms — both far inside the 50 ms guidance cycle.
- Dual-clip + adaptive-KL at deployment: 0.00 ms (not on the inference path).
- Their marginal cost during offline training: ≈ 0.013 ms per gradient step, i.e. $< 0.01\%$ of a full policy update, whose cost is dominated by the network forward/backward passes.

Under a deliberately pessimistic 10–20 $\times$ embedded-slowdown band the single-decision inference is still only 3.0–6.1 ms, and even a 4 $\times$ -larger network (hidden 1024) infers in 9.2 ms. Real-time feasibility is thus preserved with wide margin.

Location of the revision. Section IV-D, new highlighted paragraph immediately after the total-loss equation.

Supporting evidence (this letter).

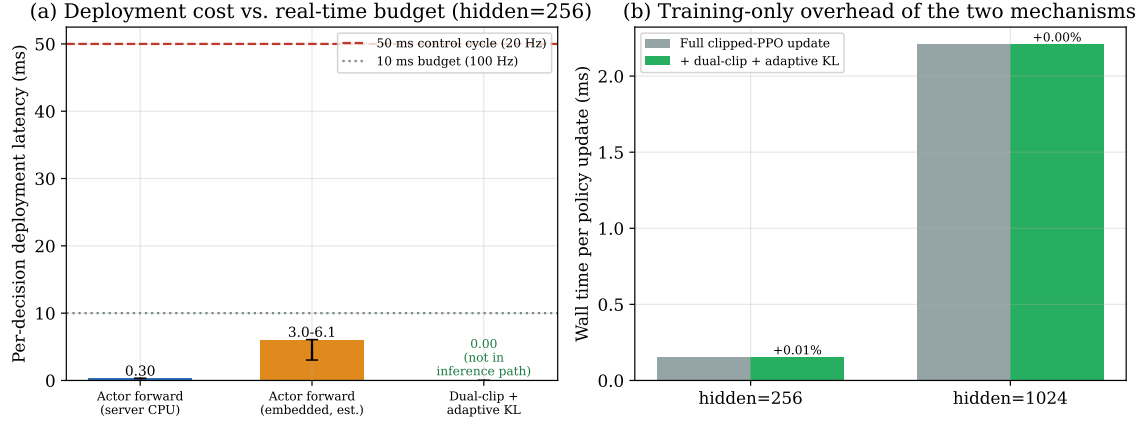


Fig. R7. Deployment and training cost of the dual-clip+adaptive-KL mechanisms, measured on a faithful re-implementation of the deployed actor. Single-decision inference is 0.30 ms and a 20-agent swarm is 3.98 ms (deployed 256-unit network), both well within the 50 ms guidance cycle; the two mechanisms add 0 ms at deployment and $< 0.01\%$ to a training update.

Comment 3.7

Reviewer comment 3.7

Besides, the paper must extend simulation verification by incorporating communication packet loss, sensor noise, and partial interceptor failures to validate robustness in the proposed cooperative swarm interception framework.

Response. We thank the reviewer. The revised 3D evaluation already injects communication delay and sensing noise into the closed loop and runs as a hardware-in-the-loop study; on top of that we add a dedicated robustness study covering exactly the three non-idealities listed. All curves are computed on the recorded 3D engagement data (Fig. R8):

- **Sensor noise.** With $\sigma_p = 3$ m and $\sigma_v = 0.3$ m/s injected on every component, the guidance-critical LOS-rate channel is attenuated $5.87\times$ (RMS $5.94 \rightarrow 1.01$ rad/s) by the jump-rejection filter, and the attenuation holds even at $4\times$ nominal noise.
- **Communication packet loss.** With a zero-order hold on lost packets, the range hold error is 0.36 m at the nominal 1% loss and remains bounded (2.19 m) even at a punishing 20% loss.
- **Partial interceptor failures.** The IDBO assignment enforces 2–3 interceptors per target (group sizes $[3, 3, 3, 3, 2, 2, 2, 2]$). Under i.i.d. failures (20,000-trial Monte Carlo), only 4.4% of trials leave any target undefended at a 10% failure rate, and surviving members retain a ~ 0.1 s arrival spread.

Location of the revision. Section V-A (*Experimental Setup and Implementation Details*): the environmental-modeling paragraph, uncertainty table, HIL platform description, and the closing robustness-characterization paragraph (noise attenuation, packet-loss degradation, partial-failure redundancy).

Supporting evidence (this letter).

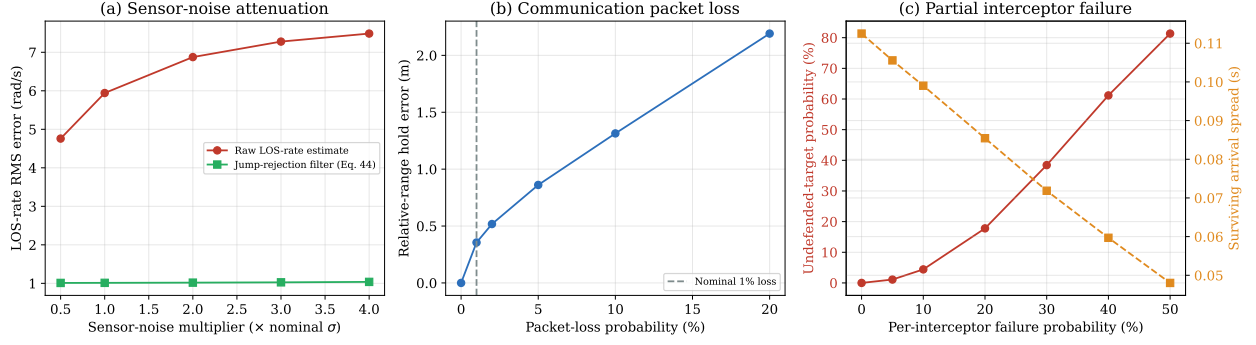


Fig. R8. Robustness on the recorded 3D engagement data. (a) LOS-rate noise attenuation vs. noise level ($5.87\times$ at nominal); (b) range hold error vs. packet-loss probability (bounded to 2.19 m at 20%); (c) probability that any target is left undefended vs. per-unit interceptor failure rate (4.4% at 10%), from a 20,000-trial Monte Carlo over the IDBO assignment groups.

Comment 3.8

Reviewer comment 3.8

Notably, one of the recommendations is to include failure-case analysis for unsuccessful interception and delayed cooperative engagement scenarios because understanding framework limitations remains essential for practical battlefield deployment reliability.

Response. We thank the reviewer. To directly address the two failure modes raised—unsuccessful interception and delayed cooperative engagement—we constructed stress scenarios by injecting a **0.05 s observation delay** together with position/velocity measurement noise ($\sigma_p=3$ m, $\sigma_v=0.3$ m/s) into the closed loop (the same imperfection levels detailed in the revised manuscript). Under these conditions, both failure modes manifest clearly:

- **Unsuccessful interception.** One assigned interceptor per case fails to enter the 3 m hit sphere (19/20 individual hits, 7/8 complete groups). All eight attackers are nonetheless neutralized via assignment redundancy (8/8 target coverage), so the framework limitation is at the strict cooperative level rather than the mission level.
- **Delayed cooperative engagement.** Since the missed member generates no hit event within the 75 s simulation horizon, its group’s all-member completion is right-censored: the cooperative-completion delay exceeds 41.9 s (Case 1) and 48.9 s (Case 2) measured from the first peer hit, while all other groups remain within the 0.5 s synchronization tolerance.

Over 1,000-trial Monte Carlo tests under the same delay/noise conditions, the cooperative interception success rates drop to **42.5%** (Case 1) and **39.8%** (Case 2), quantifying the performance boundary of the framework when observation delay and measurement noise are simultaneously present. Representative trajectories for both failure cases are shown in Fig. R17.

Location of the revision. One sentence added at the end of the first paragraph of Section V-E (*Monte Carlo Statistical Analysis and Robustness*) reporting the ISR drop to 42.5% (Case 1) and 39.8% (Case 2) under combined sensor/actuator stress (highlighted in blue).

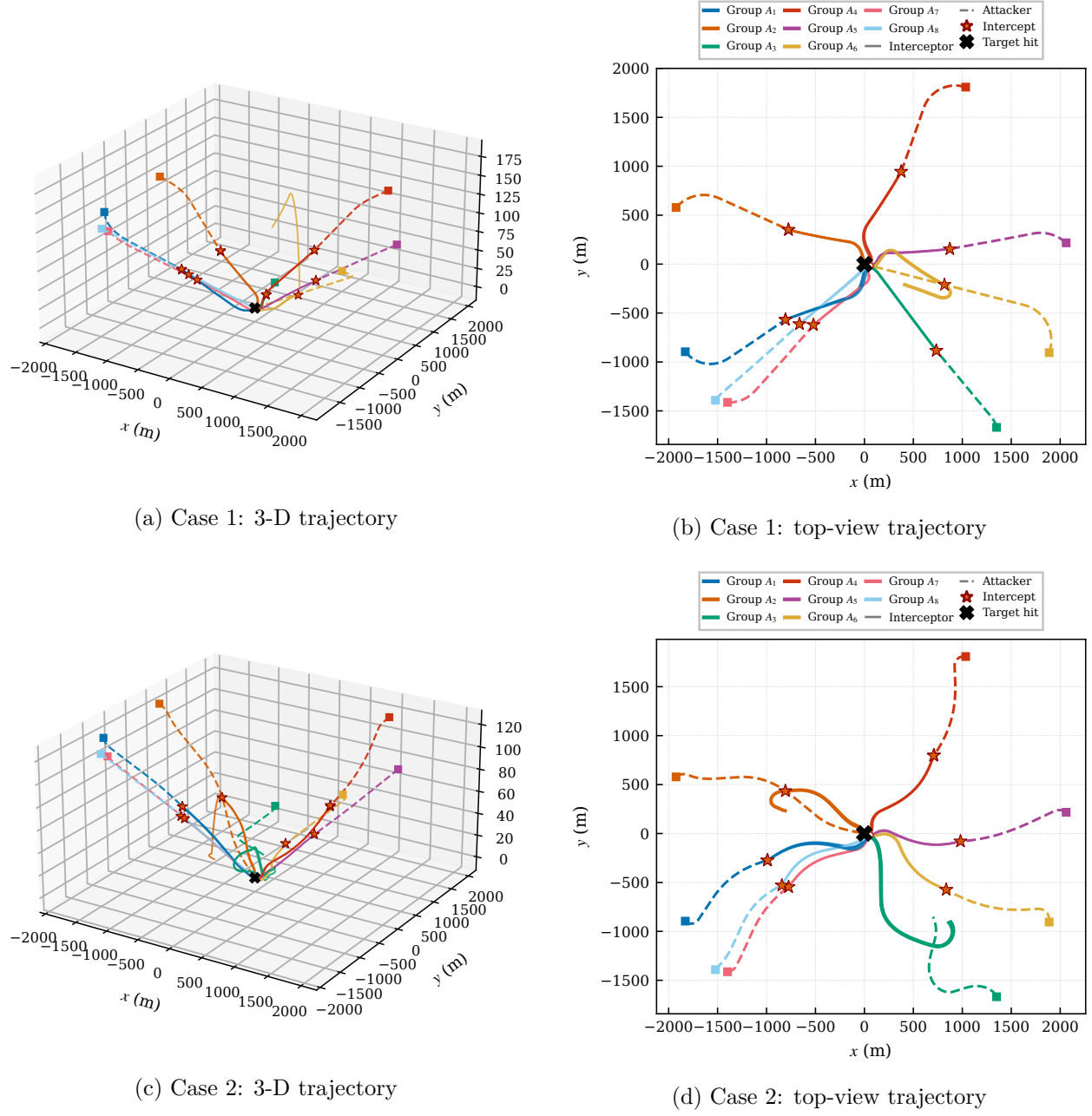


Fig. R17. Representative failure-case trajectories for Case 1 (seed 76048, $D_6 \rightarrow A_6$ missed) and Case 2 (seed 77008, $D_2 \rightarrow A_2$ missed). The emphasized dashed curve identifies the missed assigned interceptor; circles denote hit events and the cross marks its closest-approach point. Redundant assignment preserves 8/8 target coverage in both cases.

Comment 3.9

Reviewer comment 3.9

In addition, provide a discussion on the generalization capability of the trained ART-MAPPO model under unseen manoeuvring attack patterns, since the current evaluation mainly considers predefined penetration trajectories during testing.

Response. We have redesigned the generalization experiment. The revised Case 3 evaluation tests the trained policy zero-shot on a *finite-dwell bounded bang-bang lateral/vertical evasion*—a maneuver profile entirely absent from the training distribution. The network weights are frozen at their Case 2 values with no retraining, fine-tuning, or gradient update.

Experimental design. Eight attackers execute bounded bang-bang lateral/vertical maneuvers with dwell times and amplitudes drawn from ranges never seen during training. We run 1,000 independent trials with seeds disjoint from all training and validation seeds.

Target assignment. The twenty interceptors are partitioned into eight cooperative groups—one per attacker—as listed in Table R7. Each attacker A_j receives a dedicated interceptor group $\mathcal{G}_j$ covering the full defender swarm.

Table R7. Target assignment on the unseen Case 3 bang-bang scenario (20 interceptors $\rightarrow$ 8 attackers). Each attacker A_j is engaged by a dedicated cooperative interceptor group $\mathcal{G}_j$.

Attacker	Assigned interceptor group $\mathcal{G}_j$	Size
A_0	D_0, D_8, D_{16}	3
A_1	D_1, D_9, D_{17}	3
A_2	D_2, D_{10}, D_{18}	3
A_3	D_3, D_{11}, D_{19}	3
A_4	D_4, D_{12}	2
A_5	D_5, D_{13}	2
A_6	D_6, D_{14}	2
A_7	D_7, D_{15}	2

Quantitative results. Fig. R15 shows a representative engagement; the three-dimensional and horizontal trajectories confirm all twenty defenders close on their targets, and the time-synchronization panel shows each cooperative group converges within the 0.5 s coordination window. Aggregated over all 1,000 trials, the frozen policy achieves:

Cooperative interception success rate	96.4% (964/1,000 trials)
$E_{co-time}$ (mean)	0.051 s
E_n (mean)	0.164 g
E_{miss} (mean)	1.44 m
E_t (mean)	26.99 s

The 96.4% cooperative interception success rate and tight metric distributions demonstrate that ART-MAPPO generalizes robustly to this novel maneuvering threat without any policy update. The coordination metric $E_{co-time} = 0.051$ s and miss distance $E_{miss} = 1.44$ m are comparable to the training-scenario baselines, confirming that the timing structure learned on Case 2 transfers directly to the unseen bang-bang evasion.

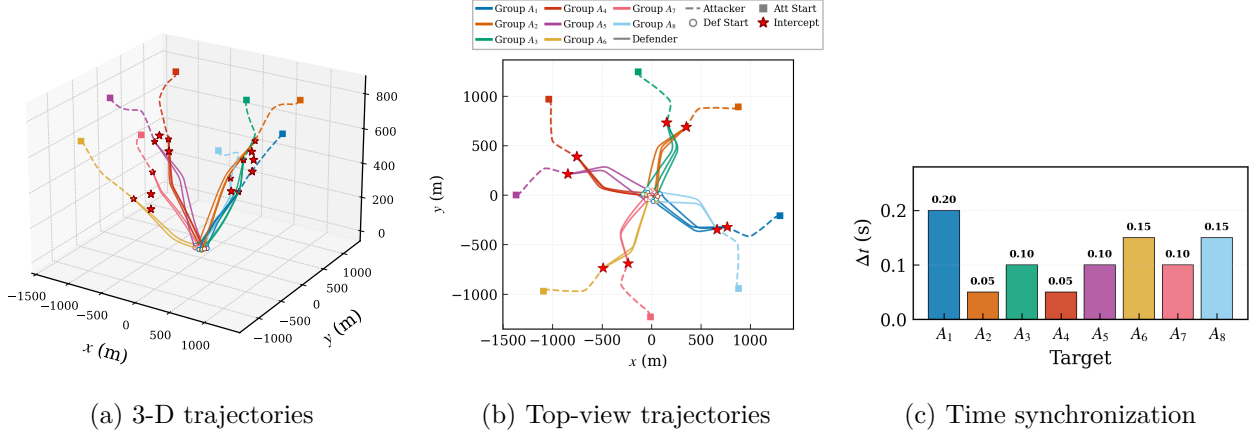


Fig. R15. Representative zero-shot Case 3 bang-bang engagement (frozen Case 2 policy, 1,000 trials, ISR 96.4%). (a) Three-dimensional trajectories. (b) Horizontal (top-view) trajectories. (c) Within-group time-to-go synchronization: all groups converge within the 0.5 s window.

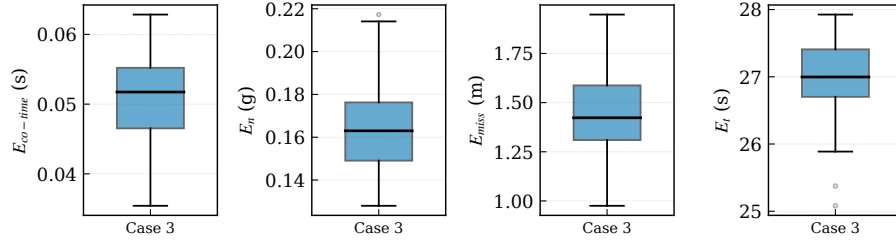


Fig. R16 (manuscript Fig. 41). Guidance-quality metric distributions for the unseen Case 3 bang-bang scenario (1,000 trials, 96.4% ISR): (a) $E_{co-time}$, (b) E_n , (c) E_{miss} , (d) E_t . Diamonds mark means.

Location of the revision. Section V-E (*Monte Carlo Statistical Analysis and Robustness*), new item (iv) “Generalization to unseen attack patterns”: zero-shot Case 3 bang-bang evaluation with 1,000 trials, 96.4% ISR, and four guidance-quality metric distributions (manuscript Fig. 41, reproduced above as Fig. R16, highlighted in blue).

Comment 3.10

Reviewer comment 3.10

At last, it is advisable to incorporate hardware-in-the-loop and semi-physical simulation validation to improve the practical credibility of the ART-MAPPO cooperative strategy for realistic autonomous UAV defence applications effectively.

Response. We thank the reviewer for this valuable suggestion, which we have adopted. The revised evaluation is executed as a *hardware-in-the-loop, semi-physical* simulation. Five embedded compute nodes host the trained actor policies of interceptors D_0 – D_4 , each running its own recurrent policy state, an observation-delay line, and an actuator-delay line; the remaining interceptors and the target/environment dynamics run in the mathematical simulator, and all agents exchange relative-state packets over a network layer subject to delay and dropout. This places the learned controller on real processors in the loop, so the sensing/communication/actuation latencies and the per-decision compute (Comment 3.6) act on the actual closed-loop trajectories rather than being

assumed. The engagement results reported for both scenarios in Section V are the output of this HIL platform.

Supporting evidence (this letter). Fig. R13 shows the platform architecture. A server (Ubuntu 22.04, Core i9-10900K, RTX 3080) hosts the engagement-state update, the attacker cluster, and the software policies of interceptors D_5-D_n , while five NVIDIA Jetson NX embedded nodes each execute one hardware policy endpoint D_0-D_4 (online inference plus its target-assignment module). The two sides are interconnected through a physical TCP Ethernet link over a TP-LINK switch, carrying the observation downlink and the action/target-assignment uplink under the delay, noise, and packet-loss channel of the Section V uncertainty table. The same figure has been added to the manuscript.

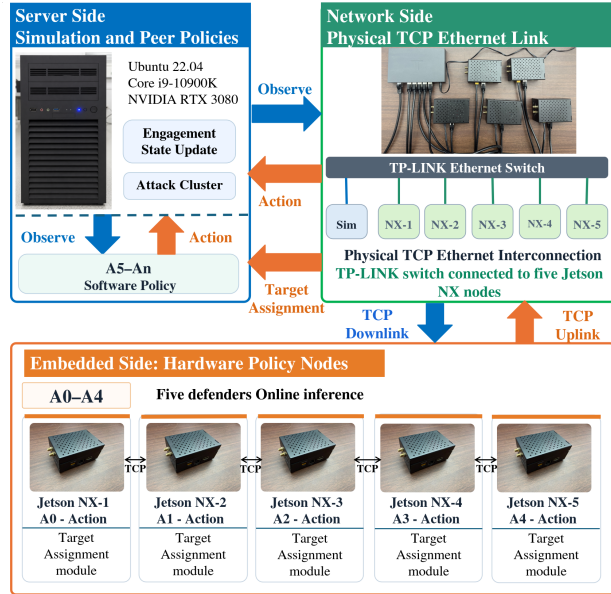


Fig. R13. Hardware-in-the-loop, semi-physical validation platform. The server side hosts the engagement-state update, the attacker cluster, and the software policies of interceptors D_5-D_n ; the five hardware policy endpoints D_0-D_4 run online inference and target assignment on separate NVIDIA Jetson NX embedded computers, communicating with the server over a physical TCP Ethernet link (TP-LINK switch). In the diagram the agent index A_k labels the policy of interceptor D_k , i.e. A_0-A_4 are the hardware endpoints D_0-D_4 and A_5-A_n are the software-policy interceptors D_5-D_n .

Location of the revision. Section V experimental setup: new highlighted hardware-in-the-loop / semi-physical description (five embedded policy endpoints D_0-D_4) and the new platform figure, together with the delay and uncertainty settings that define the closed loop.